

Lab 2 — Finding Similar Things (software)

Monday, September 28

Last updated: September 25, 2026.

Computational lab (moved from Friday). No new Unit I theory today — practice angle, nearest neighbors, and clustering ideas from Lectures 08–10 in Octave or Python (building on short [Lab 1](#)). [Quiz 4](#) was Friday, September 25. Date is on the [schedule](#).

Credit: Lab points are for effort in class, not for a Canvas file upload. Work the hub exercises; [handwrite](#) on the [one-pager](#) hard copy and get initials after a short check-in.

Resources: [Syntax sheets](#) · [Lachniet Ch. 1](#) · [Resources](#) · [Lectures 08–10](#).

What we are doing

Use software to compute many distances, name a nearest neighbor, see how feature scale changes the answer, and optionally run one k -means assignment step. [Quiz 4](#) (same skills, hand fluency) was Friday.

Learning objectives

1. Load and inspect a small dataset stored as vectors / a matrix of rows.
2. Compute many distances with Octave or Python.
3. Adapt supplied code to identify a nearest neighbor (and optionally assign clusters).
4. Validate output on a hand-checkable subset.
5. Interpret a nearest-neighbor or cluster result in context.

Math connections

Norm and distance ([Lecture 07](#)); angle / cosine ([Lecture 08](#)); nearest neighbors ([Lecture 09](#)); k -means assignment ([Lecture 10](#)); short Lab 1.

Key syntax (this lab)

Full tables: [Syntax reference](#). Essentials:

Task	Octave	Python
Norm / distance	<code>norm(x-y)</code>	<code>np.linalg.norm(x-y)</code>
Distances to rows of Z	<code>sqrt(sum((Z-x').^2, 2))</code>	<code>np.linalg.norm(Z - x, axis=1)</code>
Nearest index	<code>[~, j] = min(d) (1-based)</code>	<code>j = np.argmin(d) (0-based)</code>
Cosine similarity	<code>(a'*b)/(norm(a)*norm(b))</code>	<code>(a @ b)/(np.linalg.norm(a)*np.linalg.norm(b))</code>

Exercises

Pick Octave or Python for the term.

E1. Lab 1 recall. Query $x = (2, 1)$ and candidates as rows of

$$Z = \begin{bmatrix} 2 & 2 \\ 0 & 1 \\ 3 & 1 \\ 2 & 0 \\ 1.5 & 1.5 \end{bmatrix}.$$

Compute all five distances and the nearest index. Confirm against Lab 1 expected values (1, 2, 1, 1, $\sqrt{1/2}$; nearest z_5).

Octave:

```
x = [2; 1];
Z = [2 2; 0 1; 3 1; 2 0; 1.5 1.5];
d = sqrt(sum((Z - x').^2, 2))
 [~, j] = min(d)    % 1-based
```

Python:

```
import numpy as np
x = np.array([2.0, 1.0])
Z = np.array([[2, 2], [0, 1], [3, 1], [2, 0], [1.5, 1.5]])
d = np.linalg.norm(Z - x, axis=1)
j = np.argmin(d)    # 0-based
```

E2. Rescale. Multiply the first feature of x and every row of Z by 10. Recompute the nearest neighbor(s). Did the answer change?

Expected: tie between z_1 and z_4 ; yes, changed.

E3. Slightly larger set. Six “song” feature vectors (energy, danceability), one per row:

```
0.2  0.1
0.3  0.2
0.25 0.15
0.9  0.8
0.85 0.9
0.95 0.75
```

Query $x = (0.28, 0.18)$. Find the nearest neighbor. By eye, which group (low-energy vs high-energy) does x belong with?

E4 (stretch). One k -means assignment. Representatives $z_1 = (0.25, 0.15)$, $z_2 = (0.9, 0.8)$. Assign each of the six songs to the nearer representative. List the two clusters by row number (1-based).

E5 (stretch). Cosine vs distance. For $a = (1, 0)$ and $b = (10, 0.1)$, compute $\|a - b\|$ and the cosine similarity. In one sentence: which comparison treats them as “more similar,” and why might that matter for documents?

One-pager

Preview the [Lab 2 ticket](#); write on the hard copy in class.

After Quiz 4

[Quiz 4](#) covered Lectures 08–10. Dependence / span / basis are Midterm 1 material.